Social Networks, 'Social Death,' and Enslaved Communities in Mexico City, 1628-1637: A Provocation

First, a thought experiment – ponder the implications of Figures 1 and 2 for understandings of enslaved community formation in the Atlantic World. Both are 'social network graphs' created from 427 marriage applications for enslaved peoples in Mexico City, 1628-1637.[1] Graphs of networks, social or otherwise, are abstractions that allow us to visualize how categories of 'stuff' are interconnected in some way and to produce algorithmic or statistical analyses of those connections. That stuff is represented as *nodes* or dots on the graph and the connections as *edges* or the lines between them. Generally, digital humanists employ social networks to better understand two broad patterns: communication networks – such as letter networks or the spread of information in newspapers – or as "signatures of social identity – the patterns of relations between individuals that map the underlying preferences and characteristics of the individuals themselves."[2] Figure 1 visualizes the social relations of enslaved peoples recorded in these marriage applications. The nodes are 1554 participants identified therein (1373 of whom were enslaved) and the edges are 2140 distinct relationships articulated in the petitions – either the link between the bride and groom or the those between each and at least two witnesses who

[1] These applications preceded the reading of banns and then a marriage ceremony. Data drawn from the Archivo General de la Nación de México, Ramo de Matrimonios, volumen (volume), expediente (record). Hereinafter cited as AGNM vol:exp. All the digital analyses and data visualizations here are my own and were performed with the R statistical computing software (version 4.5.3; 2026). Claude.ai served as a coding-assistant. The data and code are available here: https://osf.io/yv9z2/overview?view_only=aec7e49acbf04cc982ef1f4f653c5393

[2] Duncan J. Watts, *Six Degrees: The Science of a Connected Age*, First edition (W.W. Norton, 2003), 48; Ruth Ahnert, Sebastian Ahnert, Catherine Nicole Coleman, and Scott B. Weingart, *The Network Turn: Changing Perspectives in the Humanities* (Cambridge University Press, 2020); and, Dan Edelstein, Paula Findlen, Giovanna Ceserani, Caroline Winterer, and Nicole Coleman, "Historical Research in a Digital Age: Reflections from the Mapping the Republic of Letters Project," *The American Historical Review* 122:2 (2017): 400-424.

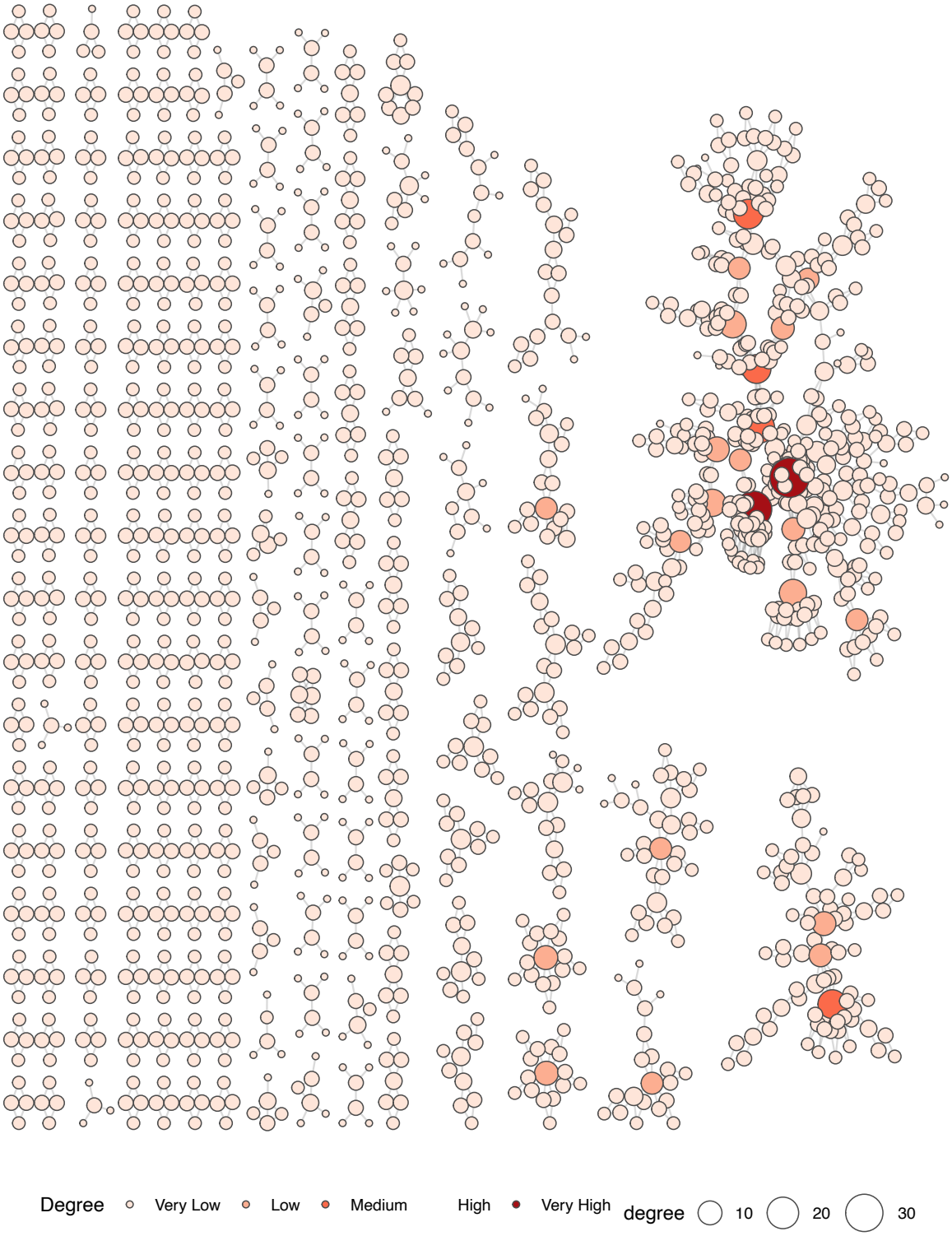


Figure 1- Enslaved Social Network, Participants from Marriage Applications, Mexico City, 1628-1637

testified regarding their eligibility to marry. When any bride, groom, or witness from one wedding appeared in another they served as a bridge linking both into a larger subnetwork.

Figure 1 is striking. Three general patterns become immediately obvious – numerous seemingly disconnected wedding parties (160 in total), 32 smallish subnetworks connecting participants from between two and ten weddings, and two larger subnetworks that include 26 and 138 interconnected marriages, representing 79 and 410 participants respectively (Table 1). But the eyes are drawn to the two extremes. The isolated subnetworks might point to the destructive atomizing power of slavery itself; the large subnetwork, conversely, might serve as evidence of enslaved people's collective negation of that condition. It is, perhaps, too tidy.

			Participants	
	Subnetworks	Weddings	All	Enslaved
Subnetworks				
Single Weddings	157	157	687	570 (43%)
2-9 Weddings	32	99	344	302 (23%)
Medium	1	24	73	73 (6%)
Large	1	129	385	372 (28%)
Total	191	409	1489	1317 (100%)

Table 1 – Information on Slave Marriage Social Networks, Mexico City, 1629-1637. Data drawn from marriage applications in the Archivo General de la Nación, México, Ramo de Matrimonios (AGNM). Full dataset available at: https://osf.io/yv9z2/overview?view_only=aec7

Figure 2 presents the exact same data as Figure 1 but with an added connection – that between individual participants and the households that enslaved them. The 215 black squares in this graph represent slaveholding households that owned two or more participants, and they serve as placeholders to connect the 563 participants who lived in such households.[3] Thus, Figure 2 approximates a *potential* enslaved social network by bringing together two different types of

[3] The 881 slaveholding-households with only one enslaved participant were excluded for visual clarity.

connection evident in the data: the affective choices of the enslaved themselves rendered visible in Figure 1 – who they married, who they called to witness – and the structural ties created by shared connections to common slaveholdings. Doing so surfaces the possibility of a single, expansive enslaved social network that includes over 8 in 10 found enslaved participants. This possibility defies easy explanation within existing frameworks. (Table 2)

<FIG. 2 NEAR HERE>

As hard as it is to process, Figure 2 is likely the more accurate representation of the communities forged by enslaved people in Mexico City in the 1630s. If true, this upends our presumptive understandings of slavery. The goal here is the use the cognitive dissonance created by considering these two graphs, and the methodological choices in creating them, to offer two interrelated provocations: that digital techniques, understood not as objective demonstrations but as humanistic experiments, open new possibilities for historical inquiry; and that current conceptions of slavery may foreclose as much as they reveal.

The first provocation, while not entirely new, requires historians to both learn new digital techniques and to expand upon the way their colleagues in the social and natural sciences treat data and data visualization.[4] Historian Jesse Hoffnung-Garskof cautions that computer-generated images carry an unearned aura of scientific authority and argues that, echoing the dean of data visualization Edward Tufte, "even in the sciences, visual representations of data typically do not prove arguments: they illustrate arguments that have been validated independently."[5] This

[4] John Thiebault, "Visualizations and Historical Arguments," in *Writing History in the Digital Age,* Kristin Nawrotzki and Jack Doughtery, eds. (University of Michigan Press, 2013): 173-185, here 183.

[5] Jesse Hoffnung-Garskof, "Cuban Racial Politics in Nineteenth-Century New York: A Critical Digital Approach," *The American Historical Review* 126, no. 1 (2021): 109-139, here 127, 138–39.

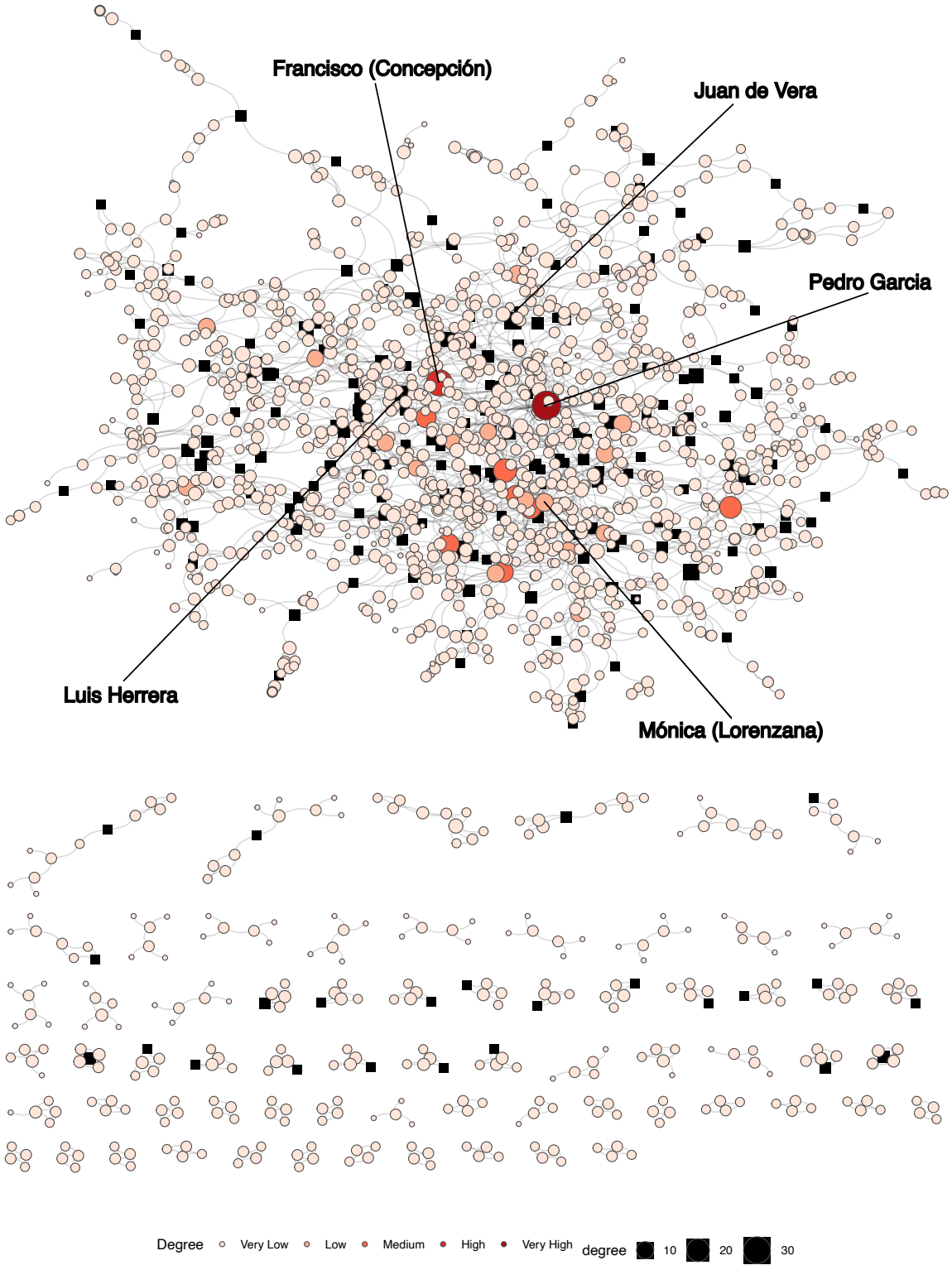


Figure 2 - Enslaved Social Network, Participants + Enslaver Households from Marriage Applications, Mexico City, 1628-1637

caution is well-founded, but insufficient. Two leading digital humanists, Johanna Drucker and Lauren Klein, point toward what lies beyond it. Rather than designing visualizations to reveal data, tell the truth about data, or illustrate an already-validated argument, they call for *humanist visualizations* designed to inspire questions, to surface the complexity and constructedness of data, and to promote sustained reflection and imaginative response. They propose we develop visualizations, in short, as provocations.[6] This reframing is more methodologically honest to the humanities, positioning digital techniques as a form of experimentation and holding visualizations to standards appropriate to humanistic inquiry on its own terms, treating them not as objective truth but as subjective artifacts.[7]

Hopefully, Figures 1 and 2 provoke. They arose from a project focused on the physical geography of slavery, an exploration of where enslaved people lived across Mexico City and the geographic patterns of their connections (with places, not people, as nodes).[8] Only later did the idea to connect individual people, not just places, and thus the first visualization, emerge. New questions and eventually hypotheses came into focus only after rendering the data in this way. Whatever patterns emerged, the job became to assess them within the intellectual contexts of the history of enslavement. This process mirrors how historians approach qualitative research by allowing historical questions and eventual arguments to flow out of the reading of the data and

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[6] Lauren Klein, "What Data Visualization Reveals: Elizabeth Palmer Peabody and the Work of Knowledge Production," *Harvard Data Science Review* 4:2 (2022): 2-3, doi: https://doi.org/10.1162/99608f92.5dec149c; and, Johanna Drucker, "Humanities Approaches to Graphical Display," *Digital Humanities Quarterly* 5:1 (2011), doi: https://doi.org/10.63744/r4ysrh7ae534

[7] According to Cameron Blevins, "Technology opens potential transformative avenues for historical discovery, but without a stronger appetite for experimentation those opportunities will go unrealized." See, "Space, Nation, and the Triumph of Region: A View of the World from Houston," *Journal of American History* 101:1 (2014), 122-147, here 147.

[8] That work is still ongoing and will be published as a companion piece to this article.

the connections and patterns found therein (recognizing, of course, that preconceptions shape those readings). The second visualization is the product of pushing further still, asking what would happen if other types of social relationships evident in the data were also included. That question, sharpened by an engagement with the scholarship on network analysis, surfaced a significant problem: despite of how easily Figure 1 fits into dominant conceptions of enslavement it is incomplete, and therefore is potentially misleading, due to missing data. The fragmentary nature of the archive serves here to reinforce preexisting expectations about slavery, and this is precisely what makes the cognitive dissonance Figure 2 produces useful.

The second provocation, also not entirely new, is to rethink our approach to slavery itself. Both potential readings of Figure 1 – crushing oppression or resistance – and the difficulty of accounting for Figure 2 at all, find their logic in competing visions of slavery highlighted in a renewed and expanded understanding of Orlando Patterson's theory of 'social death.' Patterson argued that slavery was, at its core, a system designed to violently deny the enslaved meaningful social bonds and personhood.[9] This formulation has since been adopted in far more expansive ways, becoming a dominant lens through which scholars approach Atlantic slavery.[10] For those drawn to "anatomies of doom," violence and social abjection have become the presumed norms of enslaved experience, with social death as the extreme expression of that condition.[11] For those

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[9] For the classic expression of slavery as social death, see Orlando Patterson, *Slavery and Social Death: A Comparative Study* (Cambridge University Press, 1982).

[10] It is worth distinguishing those scholars that take up social death in the process of enslavement in Africa and the Middle Passage from those who focus on slavery in the Americas. For examples of the former, see Stephanie Smallwood, *Saltwater Slavery: A Middle Passage from Africa to American Diaspora* (Harvard University Press, 2008); and James H. Sweet, "Defying Social Death: The Multiple Configurations of African Slave Family in the Atlantic World," *The William and Mary Quarterly* 70:2 (2013): 251-72.

[11] See, for example, Marisa J. Fuentes, *Dispossessed Lives: Enslaved Women, Violence, and the Archive* (University of Pennsylvania Press, 2016), 11. This tendency reaches its furthest extension in afropessimist scholarship, which applies social death as a permanent rather than

working in the tradition of "hopeful stories of heroic subalterns," social death functions less as a description of enslaved experience than as either a slaveholding ideology or a “metaphysical threat” that acts of community-building, kinship, and solidarity are understood to negate.[12] These framings keep us stuck in a familiar series of dichotomies – oppression/resistance, slavery/freedom, and now social death/social life – that historians have been grappling with, and perhaps trying to escape, since the 1950s.[13] Importantly, however, none of these articulations can entertain, let alone fully account for, the patterns in Figure 2.[14]

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historical condition; see Frank B. Wilderson, *Afropessimism*, (Liveright Publishing Corp, 2020). This position also seems to underwrite the arguments of those scholars focused the “archive as violence” metanarrative inspired by Saidiya Hartman’s “Venus in Two Acts,” *Small Axe* (2008): 176-83. Fuentes argues, for example, that “[enslaved women] appear as historical subjects through the form and content of archival documents in the manner in which they lived: spectacularly violated, objectified, disposable, hypersexualized, and silenced.” *Dispossessed Lives, 5*. Also, see Jennifer L. Morgan, *Reckoning with Slavery: Gender, Kinship, and Capitalism in the Early Black Atlantic* (Duke University Press, 2021), and the various entries in the 2016 issue of *History of the Present* devoted to Hartman’s work. For powerful rejoinders to this position see, Thavolia Glymph, “Paper Tracings in the Spectacularly Boisterous Archive of Slavery,” *The American Historical Review* 130:1 (2025): 1-18; and, James H. Sweet, “Archival Nihilism and the Methods of Revolt: Construction a New African-Atlantic Archive,” in *Africa on the Move*, Martin Löschnigg, Kristen Rüther, and Walderman Zacharasiewicz, eds., (Austrian Academy of Sciences, *forthcoming*).

[12] Herman Bennett, for example, argues that the personal relationships evident in these very wedding applications ‘steadily tempered the effects of social death.’ *Colonial Blackness: A History of Afro-Mexico* (Indiana Univ. Press, 2009), 34; Also see, Michael Lawrence Dickinson, *Almost Dead: Slavery and Social Rebirth in the Black Urban Atlantic, 1680-1807*, (University of Georgia Press, 2022); Renée Soulodre-La France, “Socially Not So Dead! Slave Identities in Bourbon Nueva Granada,” *Colonial Latin American Review* 10, no. 1 (2001): 87–103.

[13] Christopher Freeburg, *Counterlife: Slavery after Resistance and Social Death* (Duke University Press, 2021), 2; Freeburg advances the compelling argument that theories of social death are of a piece to the widely discredited arguments about the totalizing oppression of enslavement found in Stanley M Elkins, *Slavery: A Problem in American Institutional and Intellectual Life* (University of Chicago Press, 1959).

[14] Quotes drawn from Vincent Brown, “Social Death and Political Life in the Study of Slavery,” *The American Historical Review* 114, no. 5 (2009): 1231-1249, here 1235. Brown proposes treating social death as a metaphysical threat that motivated enslaved agency in his attempt to reconcile the seemingly incompatible approaches focused on doom and hope. In the end, this formulation largely situates social death as the thing being resisted.

Walter Johnson contends that in organizing our understanding of enslaved lives solely around slavery itself, whether as the experience of enslavement or the target of resistance, we risk reducing those lives to nothing more than the very institution that sought to define them. He counters that the lives of the enslaved were profoundly conditioned by, but are not reducible to, their enslavement and that the search for moments of self-determination conflates resistance with humanity. He proposes an entirely different starting point – the enslaved were fully human, and their histories encompassed not only their suffering and resistance but also the ways they lived fully within bondage, finding meaning, connection, and even love on their own terms rather than as a reflection, or negation, of their condition.[15] Humanity, in this framing, is neither something denied nor something to be recovered. Rather, it is the premise from which inquiry begins. We need to explore all the ways that experience took shape in slavery, with all its inherent inhumanness, even when it seems to contradict our assumptions. That premise, and the very different questions it opens, are what the patterns visible in Figures 1 and 2 provoke us to consider. Figure 2 simply asks the questions – what happens if we treat living in a common slaveholding as an indicator of a potential shared social universe? What might those connections add to our understandings of the social worlds inhabited by enslaved peoples in early seventeenth-century Mexico City? Does considering them help us appreciate the fuller humanness of our subjects?

The result feels radical. Part of the surprise might come from snap judgments about Figure 1. While the seemingly disconnected marriages appear to dominate the graph, most found

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[15] Walter Johnson, "On Agency," *Journal of Social History* 37, no. 1 (2003): 113-124, here 115. Elsewhere Johnson suggests that in our rush to celebrate the humanity of slaves we inadvertently treat Black humanity as probationary, something open for debate, see "To Remake the World: Slavery, Racial Capitalism, and Justice," *Boston Review*, February 1, 2017, https://www.bostonreview.net/forum/walter-johnson-to-remake-the-world/.

enslaved participants (58%) appeared in subnetworks of at least two marriages. Going further, Figure 2 presents an entirely different potential reality, one in which enslaved peoples existed within a seemingly highly interconnected, expansive community. It is likely the better representation of enslaved community in Mexico City, not because of any meaningful connections between enslavers and the enslaved, but because a focus on 'enslaved humanity' and the use of social network theory indicates that it better approximates the social worlds created by the enslaved by identifying multiple, overlapping fields of interaction.

			Node Type Breakdown		
	Networks	Nodes	Enslaved	Free(d)	Households w/ 2+ Slaves*
Large Network	1	1352	1081 (82%)	85	186
Smaller Networks	65	347	236 (18%)	87	24
Totals	66	1699	1317	172	210

Table 2 - Information on Slave Marriage Social Networks + Households, Mexico City, 1629-1637. * Represented as black squares on the graph. Data drawn from marriage applications in the Archivo General de la Nación, México, Ramo de Matrimonios (AGNM). Full dataset available at: https://osf.io/yv9z2/overview?view_only=aec7

Chronological and colonial context, as well as methodological clarity, are important to fully understand the implications of these arguments and the techniques used to develop them. The data come from a significant Spanish American urban setting in the early history of African slavery in the Atlantic World. Prior to 1650, Mexico was home to the second largest population of enslaved Africans, exceeded only by Brazil, and the largest population of free people of African descent in the Americas. Mexico City was one of, if not *the*, largest city in the Americas at the time, and was home to one of the largest urban populations of displaced Africans and their descendants in the Atlantic World. Exact population figures are scarce. An observer from 1612 speculated that 50,000 Afro-descended peoples, largely slaves, lived in Mexico City alongside

15,000 Spaniards and 80,000 indigenous people. Similarly, a modern scholar estimates that roughly 62,000 Afro-descended peoples resided in the archbishopric of Mexico, including Mexico City, compared to just over 102,000 persons of European descent in 1646.[16] Africans and their descendants thus represented a sizeable portion of the population of Mexico City before 1650. Importantly, this reality pre-existed the "sugar revolution" in Barbados in the 1640s, the transition to African slavery in the Chesapeake in the 1670s, and the subsequent explosion of sugar in the Caribbean.[17]

There are two historiographical trends that might lead us to dismiss these possibilities as exceptional before we even get started – traditional visions of urban slavery and of the distinctions between Iberian and British North American slavery. The perception that the urban setting was antithetical to slavery – shaped largely by studies of the antebellum US South after 1820 – distorts our understanding of Atlantic slavery more broadly. That perception rests on the assumption that urban slavery was a less oppressive variant than its rural counterpart because cities offered the enslaved weaker oversight from enslavers and greater freedoms relative to work and community formation.[18] Mariana Dantas argues that while numerous studies have "shown

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[16] Antonio Vásquez de Espinosa, *Compendio y descripción de las indias occidentales,* transcribed by Charles Upson Clark (Smithsonian Institution, 1948), 146; and Gonzalo Aguirre Beltrán, *La población negra de México*, 2nd ed. (Fondo de Cultura Económica, 1972), 219.
[17] Slavery in colonial Mexico followed a similar trajectory to early colonial Brazil, with the transition to Africanized plantation slavery largely complete by 1630. Stuart B. Schwartz, "Indian Labor and New World Plantations: European Demands and Indian Responses in Northeastern Brazil," *American Historical Review* 83, no. 1 (1978): 43–79; Patrick J. Carroll, *Blacks in Colonial Veracruz: Race, Ethnicity, and Regional Development* (University of Texas Press, 2001).
[18] Richard C. Wade, *Slavery in the Cities: The South, 1820-1860* (Oxford University Press, 1967); and, Claudia Dale Goldin, *Urban Slavery in the American South, 1820-1860* (Chicago University Press, 1976), 2. For studies that consider the complicated realities of urban slavery, see Frederick P. Bowser, *The African Slave in Colonial Peru, 1524-1650* (Stanford University Press, 1974), Mary C. Karash, *Slave Life in Rio de Janeiro, 1808-1850* (Princeton University Press, 1987), Pedro L.V. Welch, *Slave Society in the City: Bridgetown Barbados, 1680-1834* (Ian

that persons of African origin and descent had more opportunities (especially hiring out and manumission) to improve their economic standing within urban…environments," we need to treat these realities as integral to the operation of slavery rather than ameliorative 'breaches' in the system. These 'opportunities' could also be transformed into modes of social control and there is a preponderance of evidence that urban slavery was predicated on familiar forms of physical and psychological suffering and deprivation.[19] Rather than understand rural plantation slavery as the norm, and its urban counterpart as a less totalizing aberration, we should see the institution as a highly flexible system of exploitation in which rural and urban slavery coexisted as mutually constitutive throughout its long history in the Atlantic World.

Similarly, for some readers, marriage for enslaved peoples will come as a surprise, but for others it will serve to reinforce the "Tannenbaum" thesis that the Spanish and Portuguese legal and religious traditions endowed enslaved peoples with a legal and moral personality that served to ameliorate the worst impulses of slavery.[20] This again overdetermines our conclusions. These marriages are not simply evidence of a kinder, gentler slavery in Spanish America, but windows into how slavery operated and how enslaved people lived their lives.

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Randle Publishers, 2003), Pablo Miguel Sierra Silva, *Urban Slavery in Colonial Mexico: Puebla de los Ángeles, 1531-1706* (Cambridge Univ. Press, 2018), and María Elisa Velázquez, *Mujeres de origen africáno en la capital novohipana, siglos XVII y XVIII* (UNAM, 2006).

[19] Mary C. Karasch, "Anastácia and the Slave Women of Rio de Janeiro," in *Africans in Bondage*, ed. P. E. Lovejoy (University of Wisconsin Press, 1986), 79-105; Fuentes, *Dispossessed Lives*, 8.

[20] Frank Tannenbaum, *Slave and Citizen: The Negro in the Americas* (1946; repr., Beacon Press, 1992); For a deeper understandings of the contours of the debate surrounding the Tannenbaum thesis, see Alejandro de la Fuente, "Slave Law and Claims-Making in Cuba: The Tannenbaum Debate Revisited," *Law and History Review* 22, no. 2 (July 2004): 339–69; and the convincing rejoinder by María Elena Díaz, "Beyond Tannenbaum," *Law and History Review* 22, no. 2 (July 2004): 371–76.

Our understandings of Blacks in colonial Mexico come from often contradictory sources. A tension arises between official pronouncements from colonial authorities that clearly identified Blacks and Blackness as perceived threats to social order and an archival record replete with evidence of Black people integrating themselves into, while not being subsumed by, colonial society. They were forming families and friendships, working and attending Mass, and participating in many of the same social structures and institutions as did other colonials.

The first decades of the seventeenth century were quite tumultuous times in Mexico City, especially for Black peoples. Fears of a supposed Black rebellion rocked the city in 1612, as did a general riot in 1624 and devastating floods in 1629.[21] The 1612 'rebellion' represents a moment of state-sponsored violence that dominates modern understandings of the status of Afro-descended people in colonial Mexico. On May 2, thirty-five Black and Mulatto (mixed race of African descent) men and women were executed on the main plaza in the wake of the purported rebellion. Scholars have increasingly called into the question the validity of these specific claims of Black rebelliousness and treat them as a culmination of a palpable Spanish fear of Blackness and Black peoples.[22]

It is important to distinguish between the precipitating event – the beating death of an unnamed enslaved woman by her enslaver – and the reports of a subsequent rebellion. According to an anonymous Spanish report, as many as 1,500 Blacks seized the dead woman's body at her

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[21] Richard E. Boyer, *La gran inundación-vida y sociedad en México: 1629-1638* (Secretaría de Educación Pública, 1975); Angela Ballone, *The 1624 Tumult of Mexico in Perspective (c. 1620–1650): Authority and Conflict Resolution in the Iberian Atlantic* (Brill, 2017).
[22] María Elena Martínez, "The Black Blood of New Spain: *Limpieza de Sangre*, Racial Violence, and Gendered Power in Early Colonial Mexico," *William and Mary Quarterly, Third Series* 61, no. 3 (2004): 479–521; Daniel Nemser, "Triangulating Blackness: Mexico City, 1612," *Mexican Studies/Estudios Mexicanos* 33, no. 3 (2017): 344–66; Miguel A. Valerio, *Sovereign Joy: Afro-Mexican Kings and Queens, 1539-1640* (Cambridge University Press, 2022).

burial and paraded through the streets demanding justice for her death. Colonial authorities ignored the accusations of abuse but instead had the leaders of the protest whipped publicly and exiled from the colony. The report, and a native chronicler, gave voice to Spanish fears of Black rebelliousness prompted by this death and the subsequent crackdown. Both claimed that the Spanish believed that the supposed rebels planned to murder all Spanish men (except priests, who were to be castrated); turn all the beautiful Spanish women into sex slaves, while killing the 'old and ugly' ones; establish an African kingdom; and enslave all the natives.[23] The violence the Spanish enacted on Black bodies spoke to those fears. The victims were hung and some drawn and quartered.

In the wake of the arrests, colonial authorities also (re)issued a set of restrictions on Black peoples. They forbade Blacks from congregating in groups of more than three; limited the number of mourners at Black funerals to no more four men and four women (recall the reports of a spontaneous public outcry at a murdered woman's funeral); restricted the ability of Black women to wear jewelry and silk clothing; and, prevented Blacks from possessing weapons. This was followed in 1613 with a prohibition against Blacks and Mulattos living alone by requiring that they attach themselves to the household of a reputable Spaniard.[24] These restrictions highlight the marginalized status of Blacks in the capital and the ramifications of these events continued to shape the experiences of Black peoples decades later. When Ana sought to marry in

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[23] Luís Querol y Ruso, *Negros y mulatos de Nueva España, historia de su alzamiento de 1612* (Imprenta Hijo F. Vives Mora, 1935); Domingo Francisco de San Antón Muñón Chimalpahin Cuauhtlehuanitzin, *Annals of His Time: Don Domingo de San Antón Muñón Chimalpahin Quauhtlehuanitzin*, ed. and trans. by James Lockhart, Susan Schroeder, and Doris Namala (Stanford University Press, 2006), 221, 223.
[24] Colin A Palmer, *Slaves of the White God: Blacks in Mexico, 1570-1650* (Harvard University Press, 1976), 141.

June 1629, her witnesses testified that she had been widowed in 1612 when her previous husband, Pedro Mendoza, was among those executed.[25]

Conversely, historians have explored Black cultural life that was accepted, maybe even promoted, by colonial authorities. They identify Black confraternities or lay religious brotherhoods as key sites of Black cultural life in the early 1600s, exploring the tension between early Spanish attempts to prevent Black confraternities from forming at all and how early colonial Blacks and Mulattos used them to both build community and integrate themselves into colonial society.[26] Miguel Valerio also investigated two state-supported cultural events – a Black-led city-wide religious procession to celebrate the canonization of St Ignatius of Loyola in 1610 and a 1640 dance performance by creole Black women for the newly arrived viceroy, the highest ranking political figure in the colony, to highlight their inclusion in public life.[27]

Marriage is another important institution in such considerations. The Spanish Crown decided to promote marriage among enslaved Africans as a tool to impose social order on a

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[25] AGNM 10:126.

[26] Miguel A. Valerio, "'That There Be No Black Brotherhood': The Failed Suppression of Afro-Mexican Confraternities, 1568–1612," *Slavery & Abolition* 42, no. 2 (April 2021): 293–314; Nicole von Germeten, *Black Blood Brothers: Confraternities and Social Mobility for Afro-Mexicans* (University of Florida Press, 2006), Chap. 3. For a non-exhaustive, sample of works on Black brotherhoods in other colonial contexts, see Patricia Mulvey, "Slave Confraternities in Brazil: Their Role in Colonial Society," *The Americas* 39 (1982): 39-68; Elizabeth Kiddy, *Blacks of the Rosary: Memory and History in Minas Gerais, Brazil* (Penn State University Press, 2005); Karen B. Graubert, "'So color de una confradía': Catholic Confraternities and the Development of Afro-Peruvian Identity," *Slavery and Abolition* 33:1 (2011), 43-64; Miguel A. Valerio, "Architects of their own Humanity: Race, Devotion, and Artistic Agency in Afro-Brazilian Confraternal Churches in Eighteenth-Century Salvador and Ouro Preto," *Colonial Latin American Review* 30:2 (2021): 238-71; and the various entries in Javiera Jaque Hidalgo and Miguel A. Valerio, eds., *Indigenous and Black Confraternities in Colonial Latin America: Negotiating Status Through Religious Practices* (Amsterdam University Press, 2022); or, Cecile Fromont, ed., *Afro-Catholic Festivals in the Americas: Performance, Representation, and the Making of Black Tradition* (University of Pennsylvania Press, 2019).

[27] Valerio, *Sovereign Joy*, Chaps. 3&4.

supposedly unruly population in the sixteenth century.[28] Historians have focused on marriage to debate the extent to which Afro-descended peoples were, or were not, able to create distinct cultural communities in Mexico and beyond. Numerous studies of race, ethnicity, and identity rely on parish records like those used herein to reach divergent conclusions. They plumb individual unions and/or witness choices, in aggregate, to assess race-based community formation and the rigidity of colonial racial labels.[29]

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Overwhelmingly, considerations of community formation and racial labeling rely on measures of endogamy/exogamy, but social network analysis may provide a whole new approach. Endogamy measures focus on one relationship, but social network analysis allows us to place each union in broader webs of connections. Since the 1990s, this emerging field of interdisciplinary scholarship has argued that various types of networks – social, technological, biological, etc. – exhibit common principles. Fundamental to this approach is the notion that these networks are *not* random but share an underlying order and follow 'universal' rules that make them analyzable with mathematical tools, models, and visualization techniques.

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[28] Herman L Bennett, *Africans in Colonial Mexico: Absolutism, Christianity, and Afro-Creole Consciousness, 1570-1640* (University of Indiana Press, 2003).

[29] Dennis N Valdés, "The Decline of the Sociedad de Castas in Mexico City" (Ph.D., University of Michigan, 1978); Edgar F Love, "Marriage Patterns of Persons of African Descent in a Colonial Mexico City Parish," *Hispanic American Historical Review* 51 (1971): 79–91; R. Douglas Cope, *The Limits of Racial Domination: Plebian Society in Colonial Mexico City, 1660-1720* (University of Wisconsin Press, 1994); and Robert C. Schwaller, *Géneros de Gente in Early Colonial Mexico: Defining Racial Difference* (University of Oklahoma Press, 2016). For studies focused specifically on Afro-descended peoples see, Bennett, *Africans in Colonial Mexico,* and*, Colonial Blackness*; Colin A Palmer, "From Africa to the Americas: Ethnicity in Early Black Communities of the Americas," *Journal of World History* 6, no. 2 (1995): 223–35; and Frank T Proctor III, *"Damned Notions of Liberty": Slavery, Culture, and Power in Colonial Mexico, 1640-1769* (University of New Mexico Press, 2010).

The notion of universality may be unsettling for scholars who are grounded in the notion that knowledge is contextual and subjective. Humanities data are open to interpretation and are nuanced. Transforming those realities into a single category for analysis can appear reductive by ignoring ambiguity and complexity, the lifeblood of humanist knowledge creation.[30] Digital humanists rightly pay significant attention to the ramifications of constructing humanities datasets for computational analysis. Black Digital Humanists highlight the particular dangers in "data-ifying" Black lives. Jessica Marie Johnson argues that the origins of the idea of data as an "independent and objective statistical fact" grew out the "devastating thingification of black women, children, and men" in the context of the Atlantic slave trade. She urges scholars to avoid the pitfalls of reinscribing that practice on Black peoples and dehumanizing them again in the process. She encourages scholars to consciously deploy digital techniques to "dismantle the residue of commodification that is slavery's legacy in the New World…and [to] counteract slavery's dehumanizing impulses."[31] Social network analysis and visualizations may allow us to do just that – use data on enslaved peoples to explore the human realities that shaped their lives.

Computational analysis requires standardized data, which leads some to worry that the work of preparing or 'cleaning' data injects too much subjectivity.[32] But dataset construction,

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[30] Scott B. Weingart, "Demystifying Networks, Parts I & II," *Journal of Digital Humanities*, n.d., para. 19, accessed June 29, 2025, https://journalofdigitalhumanities.org/1-1/demystifying-networks-by-scott-weingart/.

[31] Jessica Marie Johnson, "Markup Bodies: Black [Life] Studies and Slavery [Death] Studies at the Digital Crossroads," *Social Text* 36, no. 4 (137) (December 2018): 57-79, here 58, 65-6. Also see, Kim Gallon, "Making a Case for the Black Digital Humanities," in *Debates in the Digital Humanities 2016*, ed. Matthew K. Gold and Lauren F. Klein (University of Minnesota Press, 2016), 42–49. Likewise, Jennifer Morgan worries that a desire for the "evidentiary verifiability" of quantitative data in considerations of the Middle Passage, and perhaps by extension enslavement in the Atlantic, renders more qualitative assessments secondary or tertiary in the estimation of others, see "Accounting for 'The Most Excruciating Torment': Gender, Slavery, and Trans-Atlantic Passages," *History of the Present* 6, no. 2 (2016): 184-207, here 202.

[32] Hoffnung-Garskof, "Cuban Racial Politics," 125.

qualitative or quantitative, always requires decisions that introduce subjectivity. The problem lies less in data preparation itself than in two assumptions associated with it. The first is that some types of data are inherently more objective than others.[33] The second is that cleaning implies stripping away the messiness of data to reveal a supposed underlying order. In both cases, the very term 'cleaning' is misleading because it fails to convey the thoughtful intellectual labor that should be involved when preparing humanities data for computational analysis. The researcher is making something – creating a structure, imposing an organizational logic, deciding what counts and what doesn't, determining how to handle ambiguities and lacunae, etc., just as they would with qualitative sources.[34] Rather than treating prepared data as less trustworthy, it would be more productive to recognize that all datasets are constructed, to approach them with the same critical eye brought to qualitative sources, and to be more frank about the decisions in preparing them.

An important step in preparing this data for analysis was assigning an ID to each participant and enslaver and subsequently identifying individuals who appeared in the dataset more than once. Naming practices complicate this work. A large majority of the enslaved did not list surnames, and the few that exist were undistinctive.[35] Among enslavers, variations of

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[33] Lisa Gitelman and Virginia Jackson, "Introduction," in *"Raw Data" Is an Oxymoron*, ed. Lisa Gitelman (MIT Press, 2013), 2; Catherine D'Ignazio and Lauren F. Klein, *Data Feminism* (MIT Press, 2020).

[34] Ahnert et al., *The Network Turn*, 54; Hoffnung-Garskof, "Cuban Racial Politics," 125; Katie Rawson and Trevor Muñoz, "Against Cleaning," in *Debates in the Digital Humanities 2019*, Mathew Gold and Laura Klein, eds., (Univ. of Minnesota Press, 2019), Chap. 23; and, Fred Gibbs and Trevor Owens, "The Hermeneutics of Data and Historical Writing," in *Writing History in the Digital Age*, 159-70. Casey Marina Lurtz encourages us to also "untidy our data and return to raw responses alongside summary statistics" to explore the frictions between those two forms of evidence. See, "Challenging Abstraction: Unruly Statistics and the State in Progress" *The American Historical Review* 130:1 (2025), 80-111, here 82

[35] Cope argues that many non-Spaniards in seventeenth-century Mexico City had no "official" last name until required to choose one. See, *Limits*, 62–63.

spellings and the inconsistent inclusion of maternal surnames further complicate matters. Thus, assigning IDs relied on a combination of information about the participants – name, ethnicity or race, enslaver, generalized age, occupation, place of residence, and other clues in the applications, when available. Only participants owned by a single enslaver were considered as potential repeats due to the impossibility of tracking changes in ownership through sale, inheritance, or manumission.[36]

Participants with common first names and the same enslaver required individualized decisions when assigning an ID. Erring on the side of caution, any two entries that could reasonably represent distinct individuals are coded as such. The most challenging example comes from the household of Luis Herrera which included a mid-40s Angola named Bartolomé; a mid-40s Kongo named Bartolomé; a mid-40s Kongo named Bartolomé de Mesa, and a mid-40s Black named Bartolomé de Mesa.[37] It is conceivable that all, or any two, were the same person.

Generally, potential repeats with two different *specific* racial or ethnic monikers are coded as distinct. Thus, the Angola and the Kongo Bartolomé (with or without a surname) received distinct IDs. The Black and Kongo Bartolomé de Mesa entries received a single ID based largely on the distinctive surname. In these records, the standard practice was to identify individuals as Black and then to provide clarifying information about their specific origins. For African-born individuals, 'Black' would be followed by an ethnic label like Kongo or Angola (e.g. "*negro de tierra de Angola*/Black from the land of Angola"). Creoles, people born outside of Africa, would generally be identified as such along with their city of origin ("*negro criollo de*

[36] Based on these criteria, 116 enslaved and 10 free/freed participants appeared more than once. In 28 instances, two individuals with the same first name and the same enslaver were treated as distinct based on differences in ethnicity, age, or participation in a common wedding.
[37] For a discussion of the subtle differences in these ethnonyms, see Proctor, *Damned Notions*, 46-56.

Havana/creole Black from Havana"). Despite the omission of the ethnic qualifier here, the surname and age suggested the 'Black' and 'Kongo' entries, regardless of the inclusion of the surname, were the same person. The result is two IDs, one for the Angola Bartolomé and one for the three Kongo/Black Bartolomés. A free man named Gaspar Hernandez provides another extreme case. In three different weddings, he self-identified as Angola, Black, and Mulatto. Based on the distinctiveness of his name in the dataset, and the fact that each entry included information about selling chickens on the main plaza and living in the home of a man named Oñate all three received a single ID.[38]

#

Identifying individual participants and enslavers made it possible to visualize the connections amongst them. Basically, each marriage was a combination of at least five connections; that between the bride and groom and those between each and two witnesses.[39] This results in a distinctive social network, insofar as what we are visualizing are the connections within and then among wedding parties.[40] Overwhelmingly, enslaved wedding parties consisted of four people, with two witnesses serving for both the bride and groom. This tended to result in a single marriage being represented by two overlapping triangles. For example, in the marriage of Domingo Moreno and María de la Cruz we find triangle uniting them with Pedro Garcia, and a second uniting them with Geronimo (Figure 3). These small sets of connections become

[38] AGNM 5:84-86; 1:61; 107:1; and 7:60. Variation in racial labelling was not as common as we might think. See, Cope, *Limits*, 76-7.

[39] Spiritual sponsorship like godparents at baptism and wedding witnesses is often seen as an opportunity for upward social connections, that the sponsor would be expected to be of a higher social class than those they sponsored. That does not appear to have been the case here. Witnesses were overwhelmingly enslaved, sponsoring fellow enslaved peoples.

[40] While the bonds created by marriage and those between a wedding witness and a bride or groom were not of the same magnitude, they are being treated as the same here. Any explicit connection between participants is included, and thus connections *between* witnesses are not.

integrated into larger social networks when any one of the participants appeared in another wedding. Thus, when Domingo Moreno appeared as a witness for Francisco and Ana María he connected their marriage to his own while Pedro Garcia connected it to the sixteen others that he participated in, and so on (Figures 1 & 4).[41]

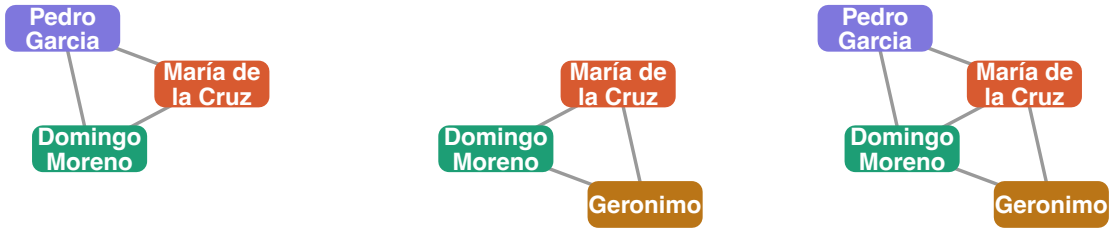


Figure 3 - Model Social Network

Figure 1 allows us to see each set of relationships found in available marriage applications, 1628-1637. Again, the two extremes evident here – the significant number of seemingly disconnected marriages and the single large sub-network – jump out. The large sub-network is eye-catching, it defies expectations and it is beautiful in many ways (Figure 4). Beautiful in appearance, it feels alive. Beautiful in that we can connect all these individual people, 410 in total, into a visible social web that we could only guess at before. Considering that marriage was not normative among the enslaved, even in Spanish America, the implications of finding 397 enslaved individuals enmeshed in a single marriage subnetwork is powerful and do, potentially, bespeak broader community formation.[42]

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[41] AGNM 161:114.

[42] Data from the later colonial period suggests that enslaved infants were baptized to unmarried parents 80-90% of the time, pointing to low marriage rates among slaves. See, Thomas Calvo,

Guadalajara y su region en el siglo XVIII: población y economía (Ayuntamiento de Guadalajara, 1992), 91; and, Proctor, *Damned Notions,* 23-5. For similar findings on Brazil see, Stuart B. Schwartz, *Sugar Plantations in the Formation of Brazilian Society: Bahia, 1550-1835* (Cambridge University Press, 1985), 389; and James H. Sweet, *Recreating Africa: Culture, Kindship, and Religion in the African-Portuguese World* (University of North Carolina Press, 2003), 36.

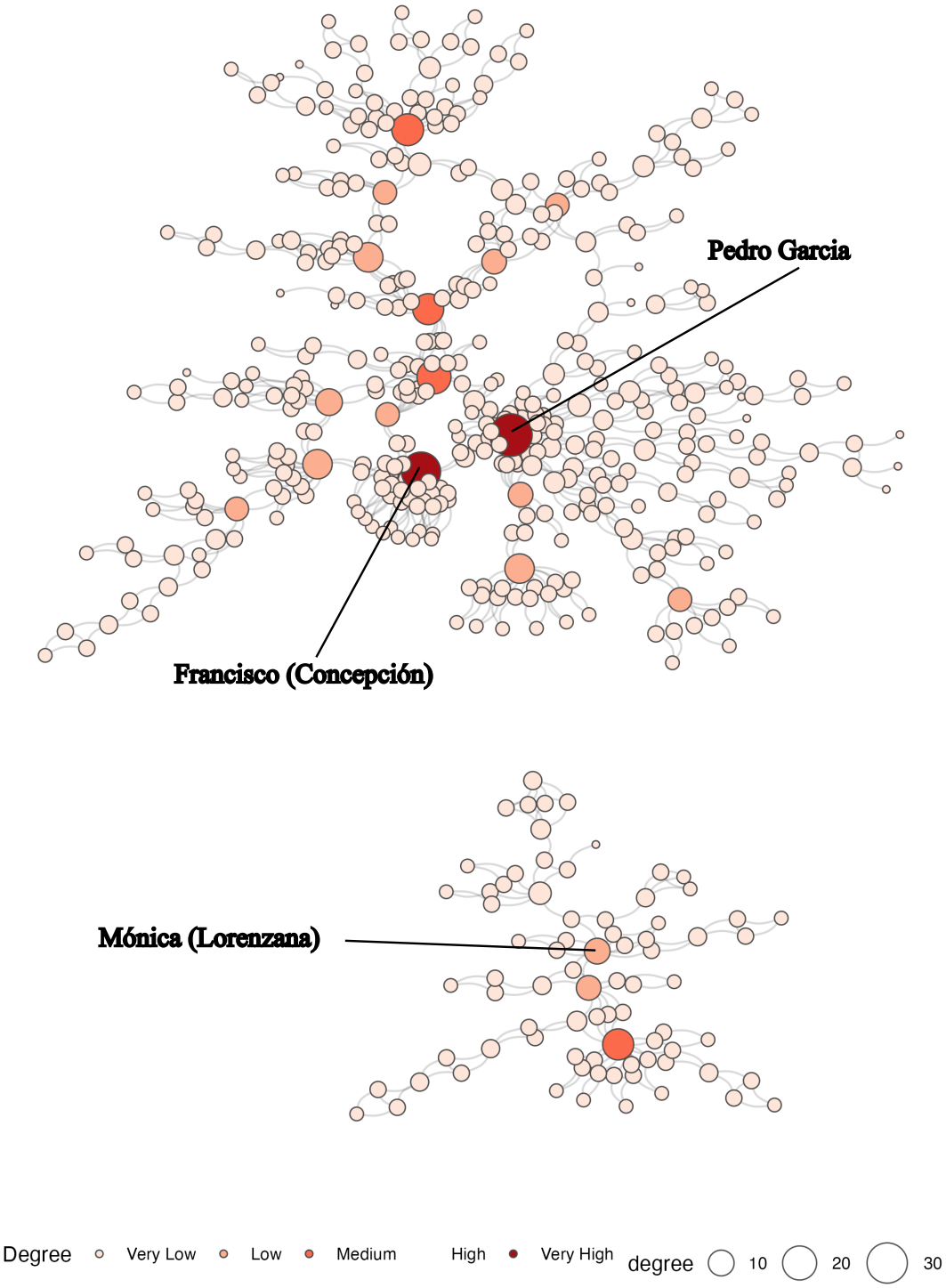


Figure 4 - Two Largest Subnetworks, Participant Social Networks, Enslaved Marriage Applications, Mexico City, 1628-1638

Figures 1 and 4 also help identify central actors within this community. For example, Pedro Garcia, the most highly connected individual in the data, served as a witness for thirty-four enslaved people in seventeen weddings. Still, we know very little about him, only what information is in the applications. He was African born, claiming an Angola ethnicity, and was in his mid-forties. He lived near the convent of Santo Domingo just a few blocks north of the Plaza Mayor and Metropolitan Cathedral. His enslaver, Captain Juan de Mansilla Hinojosa, served as the *alcalde mayor* of Taxco, which put him in charge of tax collection and the administration of the indigenous labor drafts there. Two other participants found in the dataset, Beatriz de Mansilla and Juan Garcia, also listed Mansilla as their enslaver. Juan Garcia's surname might hint that he and Pedro were related (perhaps brothers), or that they shared a previous enslaver in common.[43] We simply cannot know. A quick look at the graph suggests that even finding Pedro Garcia in the records, and counting the number of people he sponsored, cannot fully capture his place in this community.

The same is true for Francisco (Concepción) and Mónica (Lorenzana), among others. Francisco, an Angola in his mid-forties, was enslaved by the convent of Our Lady of the Conception, working in the infirmary there, on Northwestern outskirts of the city center. Perhaps he encountered the 28 people he sponsored as they sought medical care, attended funerals, or visited the black confraternity located there.[44] Mónica, a thirty-something Angola, sat at the center of the second largest network, appearing on behalf of ten people. She or her enslaver, a merchant named Francisco de Lorenzana, sold goods in the large open-air market on the main plaza. As is often the case with enslaved people, each appeared briefly in the archival record,

[43] For an example of two Kongolese brothers, see AGNM 161.51; for two creole brothers (along with their African mother), see AGNM 183.44.
[44] Valerio, *Sovereign Joy*, 61.

only to recede into the unknown. Still, these visualizations allow us to see each as important cogs in whatever communities enslaved persons created in the viceregal capital, communities that endogamy measures do not capture.

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Importantly, however, these visualizations are not representations of *the* enslaved community in 1630s Mexico City. They are better understood as abstractions, among many possible abstractions, of that community, over time, based on available data. It was not a circle of friends, but a web of individuals pulled together by affective choices, a web of potential, and potentially real, friends and acquaintances, a web that shifted over time as individuals made connections that pulled them into the network and as others disappeared from the community for various reasons.

As such, we must acknowledge that Figures 1 and 4 may not fully capture reality because data are missing, significantly impacting what can be inferred from the resulting graphs. The number of found applications for 1628-1637, while relatively dense, are wildly inconsistent. There are no found applications for the years 1630 and 1632 and annual counts range from 17 to 148 (Figure 5). Additionally, the data from 1628 are incomplete. There are as many as 200 additional applications from 1628 that remain uncollected as of this writing because the great majority of records housed in the *Ramo de Matrimonios* have been withdrawn indefinitely from public access for preservation.[45] The 17 records included here were collected prior being withdrawn. It is hard to speculate on just how much data are missing, but we can begin to approximate it. For example, the "Book of Wedding Banns" for non-Spaniards and non-Indians

[45] Personal communication from José Rodrigo Moreno Elizondo, Subdirector de Acceso y Servicios al Público, Archivo General de la Nación, México, dated July 7, 2026.

from *El Sagrario*, the most important parish in the city, lists all pending marriages to be celebrated there. The "Book" includes 781 marriages with an enslaved bride and/or groom between January 1, 1629, and April 16, 1634, for an average of approximately 142 per year.[46] Importantly, three-in-four found applications come from *El Sagrario*, with a smaller number coming from the *Santa Veracruz* and *Santa Catalina* parishes. This suggests that we are working with less than half of all potential applications for this period (even if the unavailable data from 1628 could be included).

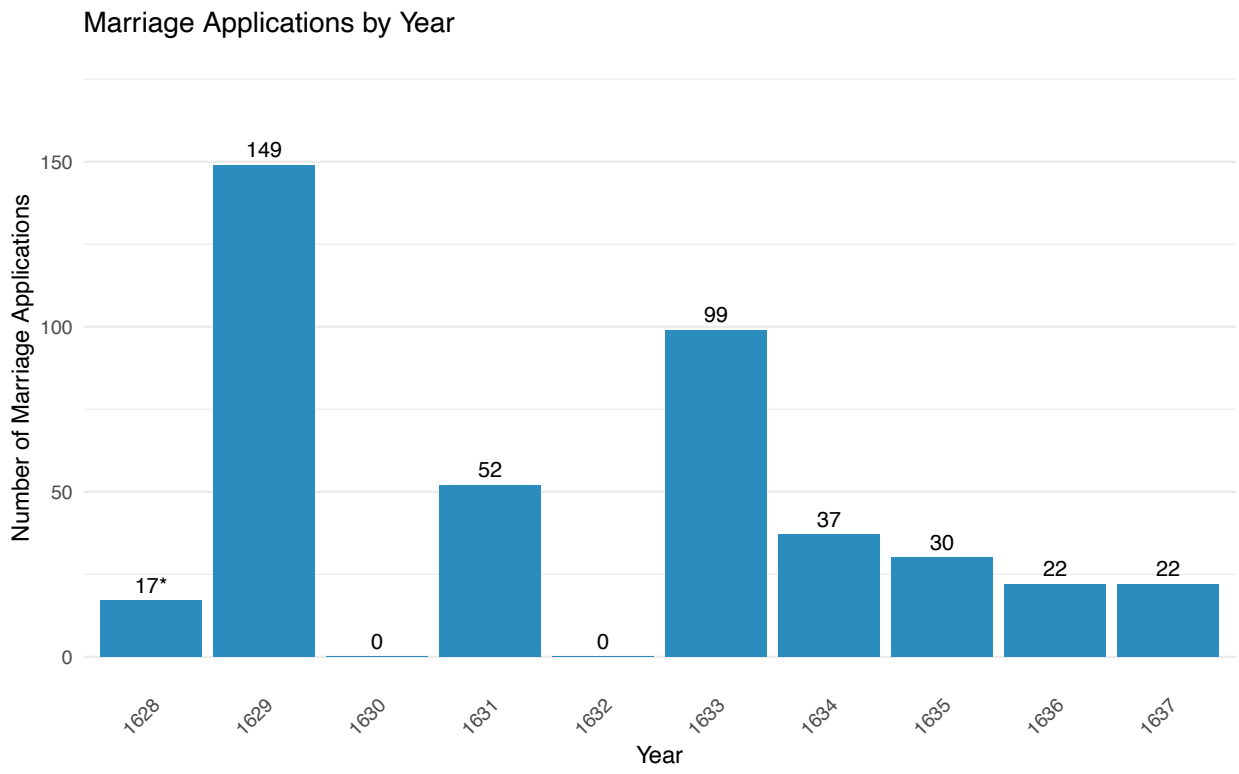


Figure 5 - Number of Extant Slave Marriage Applications, Mexico City, 1628-1637

[46] "Libro de Amonestaciones de Castas, 1629-1634," Archivo de Genealogía y Heráldica, AGN, México. Microfilm: Proyecto - DR, Sociedad - 25266, TARJ - 524, Delegación - Cuidad de México, Parroquía - Sagrario Catedral, Registro - Matrimonios de Casta y Españoles, Caja Z-D.

These records do not include data on wedding witnesses and thus cannot replace the wedding applications in building out the social network.

Network scholars can help us better understand the potential implications of that missing data. Albert-László Barabási, Réka Albert, Duncan Watts, Steven Strogatz, and others demonstrate that many real-world networks – including neural networks, the electrical power grid, the World Wide Web, scientific collaboration networks and, importantly, social networks – are organized very differently from random networks (defined by a bell-curve-like shaped distribution of connections).[47] These non-random networks consist mostly of nodes with relatively few connections and a comparatively small number of highly connected 'hubs' and thus follow a power law distribution of connectivity.[48] At the same time, social networks are typically organized into clusters of closely connected individuals, often centered on hubs, that are linked to one another by comparatively infrequent bridging relationships. Counterintuitively, these bridging or "weak" ties can play a larger role than highly connected hubs themselves in maintaining large-scale connectivity because they connect otherwise separate subcommunities.[49]

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[47] Albert-László Barabási and Réka Albert, "Emergence of Scaling in Random Networks," *Science* 286, no. 5439 (1999): 509–12; Duncan J. Watts and Steven H. Strogatz, "Collective Dynamics of 'Small-World' Networks," *Nature* 393, no. 6684 (June 1998): 440–42. For general overviews, see Mark Buchanan, *Nexus: Small Worlds and the Groundbreaking Science of Networks*, (W.W. Norton, 2002); Duncan J. Watts, *Small Worlds: The Dynamics of Networks between Order and Randomness* (Princeton University Press, 2018).

[48] To confirm that the marriage network follows a power law, I used the maximum-likelihood method proposed by Aaron Clauset, Cosma Rohilla Shalizi, and M. E. J. Newman, which replaces the older, less reliable practice of fitting a line to a log-log plot; see "Power-Law Distributions in Empirical Data," *SIAM Review* 51, no. 4 (2009): 661–703. The resulting exponent ($\alpha = 3.35$) falls within the range typically reported for real-world power-law networks, and a goodness-of-fit test confirmed the network is statistically consistent with a power-law distribution.

Random subsampling of power-law networks does not necessarily preserve the true network's degree exponent, a caution worth noting given this network's incomplete data and estimated $\alpha = 3.35$. See Michael P. H. Stumpf, Carsten Wiuf, and Robert M. May, "Subnets of Scale-Free Networks Are Not Scale-Free: Sampling Properties of Networks," *Proceedings of the National Academy of Sciences* 102, no. 12 (2005): 4221–4224.

[49] Mark S. Granovetter, "The Strength of Weak Ties," *American Journal of Sociology* 78, no. 6 (1973): 1360–80.

The present network follows these same general patterns with one important caveat. Importantly, spouses and witnesses operate almost as two distinct populations with witnesses serving as hubs and the bride/groom as bridges (97 of the 126 participants who appear more than once served only as witnesses). In our network, the marriage of Esperanza (Garcia) and Domingo (Cisneros) provides the clearest example of these realities, directly bridging the clusters centered on their witnesses Pedro Garcia and Francisco (Concepción), the two individuals with the highest connectivity.

The connectivity of some power-law networks can be highly sensitive to missing data. Whether a network appears well-connected or fragmented depends not simply on the number of relationships it contains, but on whether enough bridging ties remain to connect otherwise separate portions of the graph. Research on both theoretical and real-world networks has shown that connectivity often exhibits threshold behavior. Below a critical level, networks consist primarily of many disconnected clusters; above it, those clusters resolve rapidly into a single dominant structure known as the *giant component* (while some smaller, disconnected subnetworks remain). This abrupt shift is known as a *phase* or *percolation transition*. This process has been compared to water freezing; liquid water molecules move about independently until, at a single critical moment, they lock suddenly into the fixed structure of ice.[50] Numerous

[50] I first encountered phase transitions and the ice metaphor in Albert-László Barabási and Jennifer Frangos, *Linked: The New Science of Networks* (Basic Books, 2002), 16–17, 79, where they discuss it in random networks. Réka Albert, Hawoong Jeong, and Albert-László Barabási initially found that power-law networks are strikingly resistant to random data loss (and thus do not undergo phase transitions), a property they attributed to the presence of a small number of highly connected hubs; see "Error and Attack Tolerance of Complex Networks," *Nature* 406 (2000): 378–382. Reuven Cohen and his colleagues subsequently showed that this resistance is not a universal property of power-law networks but depends on how extreme those hubs are, formalized as a network's "degree exponent." Only networks with an exponent below approximately 3, meaning hubs disproportionate enough to make the network's connectivity effectively unbreakable, exhibit the near-total robustness that Albert, Jeong, and Barabási

studies have shown that as individuals (nodes) or relationships (edges) are removed, the giant component may remain relatively stable until that threshold is reached, after which the network undergoes a *phase transition* and contracts rapidly, fragmenting into many smaller subnetworks. The opposite is also true; a network could appear fragmented and if enough missing data was subsequently added it could eventually snap into a network dominated by a giant component.

Looking back at Figure 1, this appears to describe the pattern we observe there: a system fragmented into numerous disconnected subnetworks potentially still below its transition threshold. In this case, it is likely the loss of bridging relationships, not necessarily the disappearance of highly connected individuals, that would most directly explain the fragmentation visible in Figure 1. For example, if just one newly discovered marriage application connected a participant in the largest component (410 nodes) and the second-largest component (79 nodes), those two subnetworks would immediately merge, increasing the largest connected component to nearly one-third of all found nodes in single step. Similar connections among the remaining subnetworks would continue this process, progressively transforming an apparently fragmented network into a much more unified one.

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described. Networks with a higher exponent, including the network analyzed here ($\alpha = 3.35$), have a genuine, calculable threshold below which they can fragment. See Cohen, Keren Erez, Danile ben-Avraham, and Shlomo Havlin, "Resilience of the Internet to Random Breakdowns," *Physical Review Letters* 85, no. 21 (2000): 4626–4628. Gueorgi Kossinets confirmed this empirically, finding that a real scientific co-authorship network underwent a sharp phase transition despite possessing properties theoretically associated with robustness to random data loss; see "Effects of Missing Data in Social Networks," *Social Networks* 28, no. 3 (2006): 247–268, esp. 260–263. Jeffrey Smith and James Moody likewise found, across a sample of twelve real-world networks, that the size of a network's largest connected component remains comparatively stable until roughly 30 percent of nodes are missing, after which it declines sharply; see "Structural Effects of Network Sampling Coverage I: Nodes Missing at Random," *Social Networks* 35, no. 4 (2013): 652–668, here 661.

However, historical reconstructions such this may be especially vulnerable to missing data because the observed network reflects archival survival rather than the complete set of historical relationships. It is important to recognize that the studies cited above establishing the possibility of phase transitions simulate the effects of data loss on networks whose true extent is known. Historians working with archival sources are rarely so lucky. We do not begin with a known network and imagine its decay; we begin only with what exists, what has already decayed, with no secure inherent baseline against which to measure the loss. As such, we may not be able to argue definitively that the surviving graph lies on one side of a connectivity threshold or the other based on found data. But that does not make the resulting network's apparent shape either straightforwardly reliable or entirely uninterpretable. Here, even though the missing applications are unrecoverable, the corroborating information from the *El Sagrario* wedding banns indicates that surviving applications capture well under half of marriages. Research on phase transitions suggests fragmentation is entirely plausible when missing that much data. Consequently, Figure 1 should not be interpreted uncritically as evidence that the historical social world itself was similarly fragmented. Instead, the patterns there likely reflect the effects of missing data and substantially underestimate the connectivity of the underlying historical social network.

But an estimate of *how much* is missing does not by itself tell us what a fuller network might have looked like, or if recovering it would meaningfully change Figure 1's apparent shape. The addition of the second type of social connection evident in the data, shared slaveholding, is an attempt to rectify this problem. The logic is this – while Pedro Garcia, Francisco, and Mónica are evidence that a limited number of participants made lots of connections that anchor these networks (Figure 4), not every potential connection they could have created are evident in Figure

1. For example, Pedro Garcia, Beatriz de Mansilla, and Juan Garcia all listed the same enslaver but did not co-appear in any weddings.[51] Should we assume then that these three had no connection to each other and operated in distinct social circles?

As Figure 2 suggests, connecting participants through shared slaveholdings results in an entirely new set of possibilities, but this should not be understood as the inevitable outcome. While the result here is a single giant component, there is a counterfactual possibility in which adding connections through shared slaveholdings would appear to reinforce patterns seen in Figure 1. If the slaveholding represented firm boundaries of the social universe of individual enslaved peoples, if their spousal and witness choices were limited largely to other people within the same slaveholding, we might expect to see smaller subnetworks clearly centered on a single household or a clique of slaveholders. That is not what these connections reveal.

But let's be honest, what those new nodes literally represent are enslavers. Contemplating the enslaver-enslaved relationship as potentially shaping enslaved community formation is difficult because we struggle to understand those connections as anything other than antagonistic. Importantly, the argument is decidedly *not* that we should use this new data to chart connections among enslavers or between the enslaved and their oppressors. Nor is the goal to explore if enslavers actively dictated marriage choice, there is no evidence they did.[52] In Spanish America,

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[51] Pedro and Juan Garcia both appear in the largest subnetwork while Beatriz de Mansilla appears in a smaller subnetwork of multiple marriages.

[52] There is significant debate regarding the lengths that enslavers may have gone to control the sexuality, and specifically the reproduction, of enslaved peoples. See, Thelma Jennings, "'Us Colored Women Had to Go Through a Plenty': Sexual Exploitation of African-American Slave Women," *Journal of Women's History* 1, no. 3 (1990): 45–74; Gregory D. Smithers, *Slave Breeding: Sex, Violence, and Memory in African American History*, (University Press of Florida, 2012). Pablo Miguel Sierra Silva speculates that enslavers in the textile industry of Puebla, Mexico, sought to control the marriage patterns of their enslaved peoples to promote reproductivity; see *Urban Slavery in Colonial Mexico*.

marriage choice for slaves is understood as an expression of free will. Following the Council of Trent, all Catholics, including enslaved peoples, enjoyed the right to marry whomever the wished, free of parental or enslaver interference.[53] Enslavers received few if any immediate benefits, and theoretically faced additional restrictions on their power, when enslaved peoples married.[54] Thus, the consensus is that marriage choice reflects enslaved, not enslaver, agency.

Still, there are good reasons to question the decision to include connections to enslaver households in a visualization that seeks to understand the social worlds that the enslaved sought to create. First and foremost, the connections between a bride, her fiancée, and their wedding witnesses were simply not the same as those between her and her enslaver. The latter was a 'relationship' of extreme power, forced onto her by her very status while the former were affective choices. Moreover, even if two enslaved people suffered under the same enslaver, it is difficult to know if they overlapped with each other in time in a dataset that covers a decade. And, even if they did overlap there is simply no guarantee that two people living in the same household developed meaningful, positive, affective ties along the lines of those expressed in marriage and witness selection. Rivalry, jealousy, and generalized dislike were distinct possibilities. We see time and again that enslaved peoples were not *always* united because of their shared oppression and could actively work at cross purposes.

And yet, there are good reasons to include them. If the goal is to better understand the possible zones of interaction that shaped enslaved people's lives perhaps their connections with

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[53] This freedom began to be challenged in the eighteenth-century and was largely revoked with the issuance of the "Royal Pragmatic on Marriage" in 1785. Patricia Seed, *To Love, Honor, and Obey in Colonial Mexico: Conflicts over Marriage Choice, 1574-1821* (Stanford University Press, 1988).

[54] Alexander L. Wisnoski III, "'It Is Unjust for the Law of Marriage to Be Broken by the Law of Slavery': Married Slaves and Their Masters in early Colonial Lima," *Slavery & Abolition* 35, no. 2 (2014): 234–52.

slaveholding households do matter. Enslavers clearly shaped the social worlds our subjects inhabited, and thinking about how shared experience within the same slaveholding potentially shaped enslaved communities outside that space is a worthy exercise. The data suggest that enslaved peoples looked outside the household for spouses and witnesses – in eight in ten weddings *all* participants hailed from distinct slaveholdings. Yet, this does not diminish the importance of the household on the potential social universe of enslaved people.

The nature of social networks supports the inclusion of slaveholding households by helping us understand how new relationships tend to form. The basic unit of any social network is a dyad, the connection between two people, but the basic unit of all group structures is a triad, which arises whenever two people know a third in common. This occurs because people are much more likely to form relationships with an acquaintance of an acquaintance than some other random person, a pattern known as 'triadic closure.' Whom a person knows shapes who they are likely to come to know.[55] A shared connection to the social universe of a household means that any two individuals who lived there would be more likely to become acquainted with and form relationships with the same people than with other random people.

The data suggest that common households shaped but did not overdetermine marriage and witness selection choices. Same-household connections account for only 3.9% of all found edges (83 pairs among 2,140 edges).[56] However, participants who shared a common enslaver were more than three times more likely to appear in different weddings within the same subnetwork in Figure 1 than would be expected to occur randomly.[57] Two people who lived in

[55] Watts, *Small Worlds*, 58–61.
[56] An additional 38 pairs of participants in a common application shared a household but not a direct edge in the dataset.
[57] Participants from forty-four households who appeared in single-wedding subnetworks were excluded from this calculation, as those participants are in the same subnetwork by definition.

the same slaveholding were more likely to form relationships with people in the same social universe – not because the enslaver dictated their social ties or because that universe was restricted to that household – but because shared enslavers meant overlapping social geographies (as might a contemporary workplace or school). Thus, households structured the opportunity for relationship formation without dictating the ties themselves. Adding households as nodes and tracing connections through them rather than linking enslaved people directly maintains the general integrity of the network. Doing so visually marks the distinction between the two types of relationships in the graph while adding no more than one additional connection to each participant, pulling most of the smaller, seemingly disconnected subnetworks into a single giant cluster.[58]

Having established that adding 215 households connecting 563 participants to one another through shared enslavement surfaces a new potential reality it is worth asking how much of that connectivity was actually necessary to reorganize the network in this way. This question is best answered by setting the enslaver/household aside and asking, more simply, what is the minimum number of direct ties among participants from common slaveholdings required to pull the same 1230 participants into a giant component. Considered as direct ties between participants, rather than as the enslaver-mediated connections described above, these 215

Participants sharing a household appeared in the same marriage subnetwork 24% of the time, compared to a baseline rate of 7.5% for randomly sampled participant pairs (3.22×, $p < 0.001$, permutation test, 1,000 iterations). Including all households, the rate was 31.3% (4.19×).

[58] The results is a multi-layer graph. See, Ingeborg Van Vugt, "Using Multi-Layered Networks to Disclose Books in the Republic of Letters," *Journal of Historical Network Research* 1 (October 2017): 25-51, here 31–32; and Sébastien de Valeriola, Nicolas Ruffini-Ronzani, and Étienne Cuvelier, "Dealing with the Heterogeneity of Interpersonal Relationships in the Middle Ages. A Multi-Layer Network Approach," *Digital Medievalist* 15, no. 1 (September 2022): para. 23.

households imply 603 such pairings.[59] A minimum of 123 of those 603 pairings were structurally necessary to reconstruct the giant component. The remaining 480 pairings did not alter the network's large-scale connectivity; they reinforced ties that already existed.[60] This demonstrates just how sensitive to missing data this network is. This distribution is precisely what one would expect of a network approaching a percolation threshold: a relatively small number of bridging relationships determine whether otherwise disconnected clusters remain isolated or become incorporated into a single dominant component. Were a more complete record recoverable, it is therefore entirely plausible that the observed participant network would more closely resemble the reconstructed network shown in Figure 2.

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Importantly, adding connections to shared households functionally pulls the fractured network together, but it does not significantly alter its underlying realities. Social network analysis allows for a deeper understanding of our findings by calculating the underlying statistics of a network. Those connectivity measures demonstrate that enslaved participants were largely responsible for the size and shape of the network even after adding households as nodes. Three such measures interest us here – degree, betweenness centrality, and eigenvector centrality (Table

[59] The difference between 563 (enslaved-enslaver) and 603 (enslaved-enslaved) connections is explained by linear versus the quadradic growth of connections for each additional participant in a household. At the extreme, drawing connections between the eleven participants owned by Juan de Vera adds ten additional connections to *each* node and fifty-five total edges to the dataset.

[60] The number of households needed to supply those 123 connections depends on how efficiently the algorithm can reassemble the network. When the algorithm selects the household that joins the most still-separate subnetworks at each step, no more than 85 households suffice. When households are considered in random order, between 94 and 108 are needed (200 trials), and when considered in order of subnetworks bridged, lowest to highest, the figure is 115.

3) – and together they show that enslaved participants, not enslaver households, remain responsible for the network's size and shape even after adding households. [61]

	Degree	Betweenness Centrality	Eigenvector Centrality
1	Pedro Garcia	Pedro Garcia	Pedro Garcia
2	Francisco (Concepción)	Francisco (Concepción)	Francisco (Concepción)
3	Francisco (Nuñez)	Francisco (Nuñez)	Domingo (Cisneros)
4	Juan Francisco (Paz Cortéz)	Mónica (Lorenzana)	Esperanza (Garcia)
5	Antonio (Navarro)	Juan de Vera – enslaver	Francisca (Garcia)
6	Domingo de la Cruz (Torrijos)	Luis Herrera – enslaver	Juan (Solís)
7	Bartolomé de Mesa (Herrera)	Domingo de Santiago (Franco)	Isabel (Figueroa)
8	Francisco (Ortiz)	Bartolome (Herrera)	María (Almonici)
9	Agustin (Herrera y Orta)	Antonio (Navarro)	Domingo (Almonici)
10	Francisco (Herrera y Orta)	Manuel Salazar	Juan (Lopéz)

Table 3 – Rank by Algorithmic Measures of the Participants + Slaveholdings Social Network, Enslaved Marriage Network, Mexico City, 1629-1637. Data drawn from marriage applications in the Archivo General de la Nación, México, Ramo de Matrimonios (AGNM). Full dataset available at: https://osf.io/yv9z2/overview?view_only=aec7

Degree measures the number of direct connections for each node and helps identify hubs: Pedro Garcia and Francisco (Concepción) rank highest by this measure, a finding unchanged by the addition of enslaver households. Network analyses aren't only interested in direct connections between nodes, but also in how any two unconnected nodes are related through other nodes. Betweenness centrality measures "the likelihood that a node [is] used in the shortest 'path' from

[61] Missing data effects the specificity of these measures, but we are only interested in general trends here. See Stephen Borgatti, Kathleen Carley, and David Krackhardt, “On the Robustness of Centrality Measures Unver Conditions of Imperfect Data,” *Social Networks*, 28:2 (2006): 124-136; and, Yann Ryan and Sebastian Ahnert, “The Measure of the Archive: The Robustness of Network Analysis in Early Modern Correspondence,” *Journal of Cultural Analytics* 6:3 (2021): doi:10.22148/001c.25943.

each node in the network to every other node."[62] In other words, the algorithm calculates the shortest distance between each person and every other person in the network and assesses the relative importance of all the other intermediate nodes as a step in the shortest paths. Nodes with high betweenness appear in those shortest paths at higher rates, helping to identify individuals who bridged different regions of the network. Only the households of enslavers Juan de Vera and Luis Herrera emerge as significant here, owing to the number of participants they enslaved and/or the importance of those participants to the network. But Mónica (Lorenzana) and Bartolomé (Herrera) also emerge as key figures, despite not ranking highly in degree, because so many shortest paths passed through them. Eigenvector centrality, finally, measures a node's importance based on its connections to other important nodes. As Stephen Borgatti explains, "even if a node influences just one other node, who subsequently influences many other nodes (who themselves influence still more others), then the first node in that chain is highly influential." [63] This measure identifies individuals likely important to the social network despite not having high degree or betweenness in the available data.

Taken together, these measures suggest that enslaved people, including some women, anchored this network, even after including enslaver households. Aside from Juan de Vera and Luis Herrera, few enslavers prove statistically significant, and the patterns in Figure 2 largely

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[62] Yann Ciarán Ryan, "'More Difficult from Dublin than from Dieppe: Ireland and Britain in a European Network of Communication,'" *Media History* 24, no. 3/4 (August 2018): 458-476, here 466.

[63] Stephen P. Borgatti, "Centrality and Network Flow," *Social Networks* 27, no. 1 (2005): 55-71, here 61, as quoted in Ruth Ahnert and Sebastian E. Ahnert, "Protestant Letter Networks in the Reign of Mary I: A Quantitative Approach," *English Literary History* 82:1 (2015): 1-33, here 14. Ahnert and Ahnert also use eigenvector to identify important women in 16th-century British Protestant networks. Also see, Sébastien de Valeriola, "Can Historians Trust Centrality? Historical Network Analysis and Centrality Metrics Robustness," *Journal of Historical Network Research* 6, no. 1 (2021): 85–125.

mirror those from Figure 1. Betweenness and eigenvector measures reveal the importance of women that degree alone misses. Mónica ranks highly in betweenness, while Esperanza, Francisca, Isabel, and María all rank highly in eigenvector centrality, largely through serving as a bridge between Pedro Garcia and other significant clusters. These women, less likely to appear in multiple marriages because men dominated among witnesses 4 to 1, nonetheless proved central to the network, as their marriages linked its clusters together.

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Taking a step back, put simply, current understandings of slavery, particularly deployments of social death, cannot help us explain the patterns evident in Figure 2 even as the power of that idea makes these findings difficult to digest. In the absence of more complete data, which would likely reveal a more highly integrated network than that seen in Figure 2, it serves as a visualization of what the network *might* have looked like. If it is a fair representation of the social network, the extended community, of enslaved peoples, it feels unexplainable at first. Earlier we noted Spanish fears of Black people that exploded into violence in 1612, one suspects that if they had known that this was a potential reality that fear would have increased exponentially. Or, perhaps they did know, and that helps explain their violent overreaction. And yet, these connections emerged in the aftermath of that repression. They coexisted alongside all the violence, deprivation, and despair that defined slavery.

The goal here is not to rehabilitate slavery, but rather to reorient our approach. Much ink has been spilt trying to understand the institution from the perspective of those who suffered in it, but can we really do that when we begin from the assumption of social death? Starting there leads us to expect elements of the first visualization and to be satisfied with it as an apparent demonstration of the socially destructive realities of enslavement and/or as providing evidence of

the ‘heroic subalterns’ we hope to find. So, the real issue might be our expectations. The levels of interconnectivity, the fact that this social network looks like so many others not associated with slavery is dumbfounding. But should it be? I am left wondering if this pattern is not more normative than exceptional in Atlantic World slavery? Have we been significantly downplaying the levels of extended community formation by enslaved peoples by focusing on the individual acts that demonstrate their humanity? Based on these assumptions, are we reinscribing the individualizing impulse within slavery that Patterson sought to describe? The easy move is to dismiss these findings as non-normative because of their distinctive historical context. Or we can accept this *may define* seventeenth-century Atlantic World slavery because this was the largest urban enslaved population in the biggest city in the Americas at the time. If slavery evolved into something different by the 19th century, we need to explore and explain those transformations.

Relatedly, beginning with the anticipation of the humanness of our subjects should lead us to expect the second network. It suggests that enslaved people were just that, people, who engaged in social relations as anyone might within their particular context. Perhaps then, that is where we should start our conversations about enslavement, rather than holding up the negation of that reality and performing surprise when our subjects behaved otherwise. If we approach our subjects as people first, people enmeshed in systems and relationships of power to be sure, even the most unequal of social relationships, but people nonetheless, it hopefully shifts our work. We should expect to find the formation of friendships, families, communities, true social networks, for enslaved individuals and we should work to excavate them. More importantly, we should be prepared to explain why we don’t see them if and when we don’t. If we find a specific context

that did not allow for the expression of these most human of realities, then it is incumbent upon us to articulate, rather than assume, the particularities that prevented them.[64]

The possibilities explored herein only came into focus through experimentation with computational analysis and visualization. Social network analysis made this well-trodden data readable and assessable in a whole new way.[65] That reality can be a little seductive. In the words of Scott Weingart, the social networks 'hammer' can make everything look like a 'nail,' and there is danger in that to be sure.[66] Not every historical topic can or should be analyzed using these techniques. But the provocation here is not simply about social networks, but about the broader category of digital history and digital humanities techniques. Jesse Hoffnung-Garskof suggests that "the most fundamental question in digital history [is]: to what degree does any digital method…help generate a meaningful answer to a relevant historical question?" That answer to that question is unknowable without experimentation. In this case, the outcome may reorient our understandings of enslaved community formation in the early Atlantic World.

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[64] An example of this seen in Brenda E. Stevenson, *Life in Black and White: Family and Community in the Slave South* (Oxford University Press, 1996), which serves as a rejoinder to the findings of Herbert G. Gutman, *The Black Family in Slavery and Freedom, 1750-1925* (Pantheon Books, 1976) on the ability of enslaved peoples in the US South to form families due to the particularities of slavery in the Upper South.

[65] For a study that combines data in baptismal, marriage, confirmation, and death records to explore the influence of familial relations on political power in late-colonial Buenos Aires, Argentina, using social network analyses, see Laura C del Valle and Juan M.C. Larrosa, "Familias capitulares y poder politico en el Cabildo de Buenos Aires, 1776-1810," *América Latina en la Historia Económica* 26:2 (2019), e970. DOI: 10:18232/alhe.970

[66] Weingart, "Demystifying Networks."